



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Subject: Programming with Java		Sem: III	AY: 2026-27
Assignment No: 4			
TITLE: Develop a Java application demonstrating method overloading and the use of static variables, methods in class design.			
Subject: Programming with Java			
ASSIGNMENT No : 4			
TITLE :Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.			

Problem Statement

Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Learning Objectives

To perform conversion between primitive data types and wrapper objects using Java wrapper classes.

Theory

Wrapper classes are predefined classes that encapsulate primitive data types into objects. Java provides wrapper classes such as Integer, Double, Character, and Boolean for every primitive type. They support autoboxing, unboxing, parsing, and conversion methods. Wrapper classes are widely used in Java Collections Framework and when object representation of primitive values is required.

Source code:

C:\Users> sitap > OneDrive > Desktop > java > collage ass > J WrapperClass.java > WrapperClass

```
1  public class WrapperClass {
2      public static void main(String[] args) {
3
4          // Primitive to Wrapper (Autoboxing)
5          int num = 10;
6          Integer intObj = num;
7
8          double value = 203.5;
9          Double doubleObj = value;
10
11         // Wrapper to Primitive (Unboxing)
12         int primitiveInt = intObj;
13         double primitiveDouble = doubleObj;
14
15         // String to Wrapper
16         Integer number = Integer.valueOf(s: "123");
17         Double decimal = Double.valueOf(s: "12.34");
18
19         // Wrapper to String
20         String str1 = intObj.toString();
21         String str2 = doubleObj.toString();
22
23         // Basic operations using Wrapper classes
24         int sum = intObj + primitiveInt;
25         double product = doubleObj * decimal;
26
27         System.out.println("Primitive int: " + primitiveInt);
28         System.out.println("Primitive double: " + primitiveDouble);
29
30         System.out.println("Integer object: " + intObj);
31
32         System.out.println("Integer object: " + intObj);
33         System.out.println("Double object: " + doubleObj);
34
35         System.out.println("Converted from String to Integer: " + number);
36         System.out.println("Converted from String to Double: " + decimal);
37
38         System.out.println("Integer to String: " + str1);
39         System.out.println("Double to String: " + str2);
40
41         System.out.println("Sum: " + sum);
42         System.out.println("Product: " + product);
43     }
44 }
```

Output:

```
PS C:\Users\sitap> & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '--enable-preview -java-project\bin' 'WrapperClass'
```

```
Integer object: 10  
Double object: 203.5  
Converted from String to Integer: 123  
Converted from String to Double: 12.34  
Integer to String: 10  
Double to String: 203.5  
Sum: 20  
Product: 2511.19
```

```
● PS C:\Users\sitap> date
```

```
01 August 2026 18:57:11
```

```
○ PS C:\Users\sitap> █
```

Conclusion

This program demonstrates how wrapper classes act as a bridge between primitive data types and objects in Java. User input received as a String was converted into Integer and Double wrapper objects using `valueOf()`, which were then validated to ensure no negative salary values were entered before computing the net salary. This shows how wrapper classes are essential in real-world scenarios like payroll systems, where raw text input needs to be safely converted and processed as numeric data.

Exercise

1. Create a program to convert student marks from String format to Integer and calculate total marks.

SOURCE CODE:

```
4th ass > J StudentMarks.java > StudentMarks
1  import java.util.Scanner;
2
3  public class StudentMarks {
4
5      Run | Debug
6      public static void main(String[] args) {
7          Scanner sc = new Scanner(System.in);
8
9          System.out.print(s: "Enter number of subjects: ");
10         int n = Integer.parseInt(sc.nextLine());
11
12         int total = 0;
13
14         for (int i = 1; i <= n; i++) {
15             System.out.print("Enter marks for subject " + i + ": ");
16             String input = sc.nextLine();
17             Integer marks = Integer.valueOf(input);
18             total += marks;
19         }
20
21         System.out.println("Total Marks: " + total);
22
23         sc.close();
24     }
```

Output:

```
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sitap>cd C:\Users\sitap\OneDrive\Desktop\java\4th ass

C:\Users\sitap\OneDrive\Desktop\java\4th ass>javac StudentMarks.java

C:\Users\sitap\OneDrive\Desktop\java\4th ass>java StudentMarks
Enter number of subjects: 3
Enter marks for subject 1: 89
Enter marks for subject 2: 90
Enter marks for subject 3: 67
Total Marks: 246

C:\Users\sitap\OneDrive\Desktop\java\4th ass>date
The current date is: 01/08/2026
Enter the new date: (dd-mm-yy)
```

2. Develop an **Employee Payroll System** that accepts employee IDs, basic salary, and bonus amounts from the user. Convert the entered values into wrapper objects and perform validation operations to ensure valid salary values before calculating the net salary.

```
C: > Users > sitap > OneDrive > Desktop > java > college ass > J EmployeePayrollSystem.java > ...
1  import java.util.Scanner;
2
3  public class EmployeePayrollSystem {
4
5      Run | Debug
6      public static void main(String[] args) {
7
8          Scanner sc = new Scanner(System.in);
9
10         System.out.print(s: "Enter Employee ID: ");
11         Integer empId = Integer.valueOf(sc.nextLine());
12
13         System.out.print(s: "Enter Basic Salary: ");
14         Double basicSalary = Double.valueOf(sc.nextLine());
15
16         System.out.print(s: "Enter Bonus: ");
17         Double bonus = Double.valueOf(sc.nextLine());
18
19         if (basicSalary < 0 || bonus < 0) {
20             System.out.println(x: "Invalid salary values entered.");
21         } else {
22             double netSalary = basicSalary.doubleValue() + bonus.doubleValue();
23
24             System.out.println("Employee ID: " + empId);
25             System.out.println("Net Salary: Rs. " + netSalary);
26         }
27
28         sc.close();
29     }
```

OUTPUT:

```
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sitap>cd C:\Users\sitap\OneDrive\Desktop\java\college ass
C:\Users\sitap\OneDrive\Desktop\java\college ass>javac EmployeePayrollSystem.java
C:\Users\sitap\OneDrive\Desktop\java\college ass>java EmployeePayrollSystem
Enter Employee ID: 23456
Enter Basic Salary: 45674
Enter Bonus: 353
Employee ID: 23456
Net Salary: Rs. 46027.0

C:\Users\sitap\OneDrive\Desktop\java\college ass>date
The current date is: 01/08/2026
Enter the new date: (dd-mm-yy)
```